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A conservative semi-Lagrangian method for inhomogeneous Boltzmann equation

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ABSTRACT

The Boltzmann equation is well-known for its good predictability of behaviors of rarefied gas flows, which may not be resolved enough by continuum models. In this work, we propose a high order conservative semi-Lagrangian method for the inhomogeneous Boltzmann equation of rarefied gas dynamics. We adopt a semi-Lagrangian scheme for the convection term, which enables us to avoid CFL-type restrictions on the time step. Also, we use a fast spectral method for computation of the collision operator. In order to preserve conservative quantities, we combine a high order conservative reconstruction and a weighted optimization technique. Several numerical tests illustrate the accuracy and efficiency of the proposed method.

1. Introduction

In rarefied gas dynamics, one has to take it into account the ratio between the mean free path and physical length scale of the problem under consideration. Depending on the ratio, the so-called Knudsen number, the behavior of gases can be quite different. When the Knudsen number is small, for example, on the range $0.001-0.01$, the Navier-Stokes equation can be used for describing the flow of gases. However, for rarefied gas dynamics where the Knudsen number ranges over $\mathrm{Kn} \geq 0.1$, NS equations cannot capture the correct behavior of the gas, because the assumptions at the basis of such a continuum model do not hold any more. In this regime, a kinetic model such as the Boltzmann transport equation (BTE) could be a suitable model for capturing the correct behavior of the gas. BTE reads

$$\frac{\partial f}{\partial t} + v \cdot \nabla_x f = \frac{1}{\mathrm{Kn}} Q(f),$$

where f is the velocity distribution defined on the phase space $(x, v, t) \in \mathbb{R}^{d_x} \times \mathbb{R}^{d_v} \times \mathbb{R}$ and $\nabla_x = (\partial_{x_1}, ..., \partial_{x_{d_x}})$. The Boltzmann collision operator $Q(f)$ describes the binary collision of gas molecules:

$$Q(f) = \int_{\mathbb{R}^{d_v}} \int_{\mathbb{S}^{d_v - 1}} B(|v - v_*|, \omega) \left(f'_* f' - f_* f \right) d\omega dv_* \tag{1}$$

where $\omega \in \mathbb{S}^{d_v - 1}$ denotes a unit vector on a sphere. In (1), the pre-collision velocities v, v_* and post collision velocities v' and v'_* are related to preserve momentum and energy:

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$$v + v_* = v' + v'_*, \quad |v|^2 + |v_*|^2 = |v'|^2 + |v'_*|^2$$

i.e.,

$$v' = \frac{1}{2}\left(v + v_* + |v - v_*|\omega\right), \quad v'_* = \frac{1}{2}\left(v + v_* - |v - v_*|\omega\right).$$

The operator $Q(f)$ has the following collision invariance: $1, v, |v|^2$,

$$\int_{\mathbb{R}^{d_v}} Q(f)\begin{pmatrix} 1 \\ v \\ |v|^2 \end{pmatrix} dv = \mathbf{0},$$

which leads to the conservation of mass, momentum energy. Moreover, BTE satisfies the H-theorem which states that the entropy function, defined by

$$H(f) = \int_{\mathbb{R}^{d_x}\mathbb{R}^{d_v}} f \ln f \, dv \, dx,$$

is a non-increasing function of time, i.e. $\frac{dH}{dt} \leq 0$, and the equality holds if any only if $Q(f) \equiv 0$ (see [28]). It is possible to show that this relation holds if and only if the distribution function f has the form of

$$\ln f = a|v|^2 + b \cdot v + c.$$

Here the coefficients $a, b, c \in \mathbb{R}$ are related to macroscopic variables, density ρ, mean velocity U and temperature T by

$$\rho(x,t) = \int_{\mathbb{R}^{d_v}} f \, dv, \quad \rho(x,t)U(x,t) = \int_{\mathbb{R}^{d_v}} v f \, dv, \quad d_v \rho(x,t) RT(x,t) = \int_{\mathbb{R}^{d_v}} |v - U(x,t)|^2 f \, dv, \tag{2}$$

where R denotes the gas constant. Therefore any equilibrium function has the form of local Maxwellian:

$$\mathcal{M}(f)(x,v,t) := \frac{\rho(x,t)}{(2\pi RT(x,t))^{d_v/2}} \exp\left(-\frac{|v - U(x,t)|^2}{2RT(x,t)}\right).$$

Continuum models such as Euler and Navier-Stokes can be derived from the BTE as asymptotic limits for small values of the Knudsen number through the Chapman-Enskog expansion [5].

Computing the numerical solution to the Boltzmann transport equation has been a challenging problem due to the high dimension of phase space and the complicate structure of the Boltzmann collision operator defined by a five-fold integral. In recent years, several numerical techniques have been proposed for efficient computation of the collision operator. Most of them were based on discrete velocity model and adopted the so-called spectral methods, which enables one to accurately compute the Boltzmann collision operator with a relatively small computational effort [24,22]. On the other hand, together with the development of such a spectral method, there have been progresses on the construction of high order methods (in space and time) for the BTE [12,11,21]. Depending on how the convection term is treated, we distinguish between three classes of schemes: fully Lagrangian (particle methods), Eulerian and semi-Lagrangian (SL) methods. Particle methods have the advantage of making an efficient use of phase space, providing a simple treatment of convection and allowing computation of high Mach number flows. Their drawback is that they are usually low order accurate. For this reason, we restrict ourselves to space discretization based on a regular Cartesian mesh. In case of Eulerian approach, splitting methods [11], exponential time integrators [21] and implicit-explicit methods based on Runge-Kutta [12,3] or multi-step [4] have been applied. For large and moderate values of the Knudsen number, the main time step constraint is related to the CFL restriction due to the convection term, which may affect the efficiency of Eulerian methods. This issue could be overcome by adopting semi-Lagrangian approach, which computes the solutions on grid points by integrating the equation along its characteristic curves. In [15] the semi-Lagrangian method has been coupled with a splitting technique, obtaining second order accuracy in time.

In this paper, we propose a fourth order non-splitting semi-Lagrangian method. We adopt the classical fourth order Runge-Kutta method for the time discretization and a conservative non-oscillatory reconstruction [6] which gives fifth order accuracy in space for smooth solutions. In order to prevent the loss of conservation in momentum and energy that comes from the computation of the collision operator, we have use a weighted L^2-minimization technique [1], which differs from the classical one [16]. An alternative to preserve the moments has been recently developed in the context of spectral method [23].

Semi-Lagrangian approaches for kinetic equations have been widely used in the context of Vlasov-type equations. See for example, [9,26,7,14].

The outline of the manuscript is organized as follows. In Section 2, we derive our non-splitting SL method. Section 3 is devoted to the description of conservative reconstruction. In Section 4, we recall the spectral method for the evaluation of the collision operator. Finally, in Section 5, we present several numerical results to demonstrate the performance of our method. In the last section, we draw some conclusions.

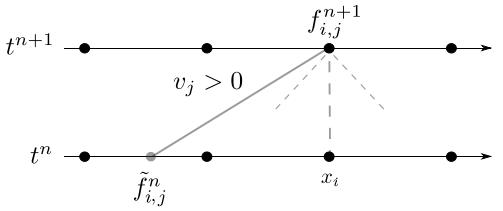


Fig. 1. Illustration of RK1.

2. Finite difference semi-Lagrangian spectral method for the BTE

2.1. Notation

Our work is based on a finite difference discretization in space and velocity. For the method, we consider a discrete velocity set

$$V = \underbrace{V_{N_v} \times \cdots \times V_{N_v}}_{d_v}, \quad V_{N_v} = \left\{ \ell \Delta v, -\frac{N_v}{2} \le \ell \le \frac{N_v}{2} - 1 \right\},$$

where Δv is a constant mesh size in velocity N_v is the total number of velocity points in each direction. Hereafter we denote the multi-index by $j \in \mathcal{J}$ to identify a specific element $v_j = (v_{j_1}, ..., v_{j_{d_v}})$ of V, where $\mathcal{J}$ is the corresponding set of indices. In this work, we focus on the problem of one-dimension in spatial domain. For the discretization in space, we use

$$\left\{ x_i = i\Delta x + x_L, \quad 0 \le i \le N_x - 1 \right\}$$

where N_x is the number of grid points in space and x_L is the leftmost point of space. The set of spatial indices will be denoted by $\mathcal{I}$. For time discretization, we use a fixed time step Δt and each time step will be denoted by

$$t_n = n\Delta t, \quad n \ge 0.$$

For each j the distribution function defined on a spatial grid point x_i at time t_n will be denoted by $f_{i,j}^n \approx f(x_i, v_j, t_n)$. Thus $f_{i,\mathcal{J}}^n$ a vector in $\mathbb{R}^{d_v \times N_v}$. When computing the macroscopic quantities, the integrals on $\mathbb{R}^{d_v}$ in (2) are replaced by discrete sums over $\mathcal{J}$ for each spatial grid x_i at time t_n, i.e.

$$\left(\rho_i^n, \rho_i^n U_i^n, E_i^n, \frac{d_v \rho_i^n R T_i^n}{2} \right) \approx \sum_j \left(1, v_j, \frac{|v_j|^2}{2}, \frac{|v_j - U^n|^2}{2} \right) f_{i,j}^n (\Delta v)^{d_v}.$$

2.2. Derivation of the first order semi-Lagrangian method

To derive the first order SL method, we begin by writing the BTE in characteristic form:

$$\begin{aligned} &\frac{df}{dt} = \frac{1}{\mathrm{Kn}} Q(f) := K \\ &\frac{dx}{dt} = v \\ &x(0) = \tilde{x}, \quad f(0, x, v) = f_0(x, v), \quad t \ge 0, \quad x, v \in \mathbb{R}. \end{aligned} \tag{3}$$

Applying the explicit Euler method to (3), we obtain

$$f^{n+1}(x, v) = f^n(x - v\Delta t, v) + \Delta t K^n(x - v\Delta t, v),$$

where $f^n(x, v) = f(x, v, t_n)$ and $K^n(x, v) = \frac{1}{Kn} Q\left(f(x, v, t_n)\right)$. Discretization in space and velocity further gives

$$f_{i,j}^{n+1} = \tilde{f}_{i,j}^n + \Delta t \tilde{K}_{i,j}^n, \tag{4}$$

where

$$\tilde{f}_{i,j}^n \approx f(x_i - v_j \Delta t, v_j, t_n), \quad \tilde{K}_{i,j}^n \approx K\left(x_i - v_j \Delta t, v_j, t_n\right).$$

Since the characteristic feet $x_i - v_j \Delta t$ could be located on an off-grid point (see Fig. 1), we need to interpolate the values of $\tilde{f}_{i,j}^n$ from $f_{I,j}^n$ and $\tilde{K}_{i,j}^n$ from $K_{I,j}^n$. For this, we consider a linear interpolation which is sufficient for the construction of first order method. For the computation of $Q_{i,j}^n$, we use the fast spectral method [22] described in Section 4. Since the macroscopic quantities except for mass are not exactly preserved, we apply the weighted L^2-minimization [1] to $Q_{i,\mathcal{J}}^n$ by imposing

$$(\Delta v)^{d_v} \sum_j Q^n_{i,j}\phi_j = 0, \quad \phi_j = 1,\, v_j,\, v_j^2.$$

Finally, we use the corrected values of $Q^n_{i,\mathcal{J}}$ to define $K^n_{i,\mathcal{J}}$, then interpolate $\tilde{K}^n_{i,j}$ and update solutions with (4). Note that the linear interpolation prevents the loss of conservation in macroscopic quantities [2].

2.3. Derivation of the high order SL methods based on RK method

In this section, we describe our high order SL methods for the BTE. First we define a backward characteristic curve to (3) which passes through the location x_i at time $t_n + c_k\Delta t$. We call this the k-th characteristic curve, and denote by $x^{(k,\ell)}_{ij} := x_i - (c_k - c_\ell)v_j\Delta t$ the location of its intersection with the line $t_n + c_\ell\Delta t$ (see Figs. 2-3). The ℓ-th stage value of the collision term $K = \frac{1}{Kn}Q(f)$ computed along the k-th characteristic corresponding to x_i and v_j will be denoted by $K^{(k,\ell)}_{i,j}$.

Applying s-stage explicit Runge-Kutta methods to (3), we obtain

$$k\text{th-stage value:} \qquad F^{(k,k)}_{ij} = F^{(k,0)}_{i,j} + \Delta t \sum_{\ell=1}^{k-1} a_{k\ell} K^{(k,\ell)}_{i,j}, \quad k = 1, ..., s,$$

$$\text{numerical solution:} \qquad f^{n+1}_{i,j} = \tilde{f}^n_{i,j} + \Delta t \sum_{k=1}^{s} b_k \tilde{K}^{(k)}_{i,j},$$

where $\tilde{K}^{(k)}_{i,j} \approx K\left(x_i - v_j(1-c_k)\Delta t, v_j, t_n + c_k\Delta t\right)$ denotes the kth Runge-Kutta flux. For each k-stage, for each i, we compute $Q^{(\ell,\ell)}_{i,\mathcal{J}}$ from $F^{(\ell,\ell)}_{i,\mathcal{J}}$ by the spectral method, then as in the first order scheme, we implement the weighted L^2-minimization so that $Q^{(k,k)}_{i,\mathcal{J}}$ are modified to satisfy

$$(\Delta v)^{d_v} \sum_j Q^{(k,k)}_{i,j}\phi_j = 0.$$

After we set $K^{(k,k)}_{i,j} = \frac{1}{Kn}Q^{(k,k)}_{i,j}$, the off diagonal elements $K^{(k,\ell)}_{i,j}$ are computed from $K^{(\ell,\ell)}_{\mathcal{I},j}$ by high order conservative reconstruction [2].

The general description of semi-Lagrangian based RK methods can be summarized as follows:

Algorithm for ERK based SL schemes

For $k = 1, ..., s$

1. Using a conservative reconstruction [6], interpolate $F^{(k,0)}_{i,j}$ on $x^{k,0}_{i,j}$ from $f^n_{\mathcal{I},j}$.
2. (Skip this for $k = 1$) For $\ell = 1, ..., k-1$, we interpolate $K^{(k,\ell)}_{i,j}$ on $x^{k,\ell}_{i,j}$ from $\{K^{(\ell,\ell)}_{i,j}\}_{i\in\mathcal{I}}$.
3. Compute:

$$F^{(k,k)}_{i,j} = F^{(k,0)}_{i,j} + \Delta t \sum_{\ell=1}^{k-1} a_{k\ell} K^{(k,\ell)}_{i,j}.$$

4. Construct $Q^{(k,k)}_{i,\cdot}$ using $\{F^{(k,k)}_{i,j}\}_{j\in\mathcal{J}}$ using the fast spectral method described in Section 3.
5. For each i, using the weighted L^2-minimization procedure described in Section 5, we modify $Q^{(k,k)}_{i,\mathcal{J}}$ so that it satisfies

$$(\Delta v)^{d_v} \sum_j Q^{(k,k)}_{i,j}\phi_j = 0, \quad \phi_j = 1, v_j, |v_j|^2/2$$

   up to a tolerance (tol).
6. Compute $K^{(k,k)}_{i,j} = \dfrac{Q^{(k,k)}_{i,j}}{Kn}$ for each i, j
7. If iterations over, update numerical solution:

$$f^{n+1}_{i,j} = \tilde{f}^n_{i,j} + \Delta t \sum_{\ell=1}^{s} b_\ell \tilde{K}^{(\ell)}_{i,j},$$

   where $\tilde{K}^{(\ell)}_{i,j} \approx K(x_i - v_j(1-c_\ell)\Delta t, v_j, t_n + c_\ell\Delta t)$.

In this work, we consider two Runge-Kutta methods for time discretization: the third order method (RK3) proposed in [27] and the classical RK4. Here we give the Butcher's tables for these methods.

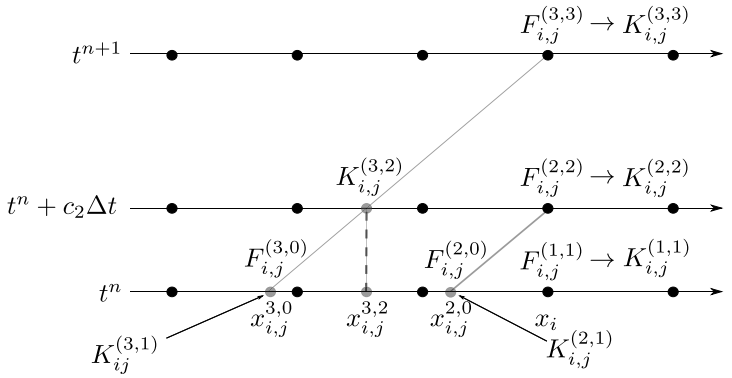


Fig. 2. Illustration of RK3 method. The quantities on grey points need interpolation.

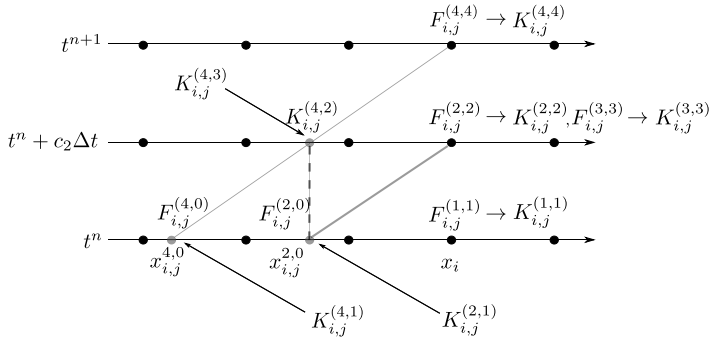


Fig. 3. Illustration of RK4. The quantities on grey points need interpolation.

$$\text{RK3} = \begin{array}{c|ccc} 0 & 0 & 0 & 0 \\ \alpha & \alpha & 0 & 0 \\ 1 & 1+\frac{1-\alpha}{\alpha(3\alpha-2)} & -\frac{1-\alpha}{\alpha(3\alpha-2)} & 0 \\ \hline & \frac{1}{2}-\frac{1}{6\alpha} & \frac{1}{6\alpha(1-\alpha)} & \frac{2-3\alpha}{6(1-\alpha)} \end{array}, \quad \text{RK4} = \begin{array}{c|cccc} 0 & 0 & 0 & 0 & 0 \\ 1/2 & 1/2 & 0 & 0 & 0 \\ 1/2 & 0 & 1/2 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 \\ \hline & \frac{1}{6} & \frac{1}{3} & \frac{1}{3} & \frac{1}{6} \end{array}$$

In both cases, $c_s = 1$ gives $\tilde{K}_{i,j}^{(k)} = K_{i,j}^{(s,k)}$, $k = 1, ..., s$, thus no interpolation is needed for the last Runge-Kutta flux. In the RK3 method, the choice $\alpha = \frac{1}{3}$ gives $b_1 = 0$, which allows us to save memory for the numerical solution. Moreover, on the RK4 method we have $a_{31} = a_{41} = a_{42} = 0$, which simplifies the computation of stage values by avoiding interpolation.

$$\begin{aligned}
\text{1st-stage value:} \quad & F_{i,j}^{(1,1)} = f_{i,j}^n \\
\text{2nd-stage value:} \quad & F_{i,j}^{(2,2)} = F_{i,j}^{(2,0)} + \frac{\Delta t}{3} K_{i,j}^{(2,1)} \\
\text{3rd-stage value:} \quad & F_{i,j}^{(3,3)} = F_{i,j}^{(3,0)} - \Delta t K_{i,j}^{(3,1)} + 2\Delta t K_{i,j}^{(3,2)} \\
\text{numerical solution:} \quad & f_{i,j}^{n+1} = F_{i,j}^{(3,0)} + \frac{3\Delta t}{4} K_{i,j}^{(3,2)} + \frac{\Delta t}{4} K_{i,j}^{(3,3)}.
\end{aligned}$$

$$\begin{aligned}
\text{1st-stage value:} \quad & F_{i,j}^{(1,1)} = f_{i,j}^n \\
\text{2nd-stage value:} \quad & F_{i,j}^{(2,2)} = F_{i,j}^{(2,0)} + \frac{\Delta t}{2} K_{i,j}^{(2,1)} \\
\text{3rd-stage value:} \quad & F_{i,j}^{(3,3)} = F_{i,j}^{(2,0)} + \frac{\Delta t}{2} K_{i,j}^{(2,2)} \\
\text{4th-stage value:} \quad & F_{i,j}^{(4,4)} = F_{i,j}^{(4,0)} + \Delta t K_{i,j}^{(4,3)} \\
\text{numerical solution:} \quad & f_{i,j}^{n+1} = F_{i,j}^{(4,0)} + \frac{\Delta t}{6} K_{i,j}^{(4,1)} + \frac{\Delta t}{3} K_{i,j}^{(4,2)} + \frac{\Delta t}{3} K_{i,j}^{(4,3)} + \frac{\Delta t}{6} K_{i,j}^{(4,4)}.
\end{aligned}$$

Remark 2.1. The RK3 based SL method involves three computations for the collision operator and five interpolations (two for distribution function and three for collision operator.) In the algorithm based on RK4, we can save computational costs by avoiding

unnecessary redundant computations. As in the RK3 method, no computation is need for computing $F_{i,j}^{(1,1)}$ on the first stage. On the third stage, we can take $F_{i,j}^{(3,0)} = F_{i,j}^{(2,0)}$ because $c_2 = c_3$. In addition, no interpolation is needed for computing $Q_{i,j}^{(3,2)}$ because $Q_{i,j}^{(3,2)} = Q_{i,j}^{(2,2)}$. When updating numerical solutions, the value of $c_4 = 1$ enables us to set $\tilde{f}_{i,j}^{n} = F_{i,j}^{(4,0)}$, $\tilde{Q}_{i,j}^{(1)} = Q_{i,j}^{(3,1)}$ and $Q_{i,j}^{(4,2)} = Q_{i,j}^{(3,2)}$. Totally, four evolutions of collision operator and six interpolations are required (two for distribution function and four for collision operator.)

2.4. Derivation of the high order SL methods based on multi-step methods

Here we present high order SL methods for the BTE based on explicit multi-step methods. In particular, we focus on the Adams-Bashforth (AB) methods. In the methods, we only need to consider one backward characteristic curve to (3) which passes through the location x_i at time t_{n+1}. The location of characteristic foot which meets the time level t_{n+1-k} is denoted by $x_{i,j}^{n,k} := x_i - kv_j\Delta t$. We define by $K_{i,j}^{n,k} \approx K(x_{i,j}^{n,k}, v_j, t_{n+1-k})$ the distribution function on $x_{i,j}^{n,k} := x_i - kv_j\Delta t$ (see Fig. 4). Applying s-step explicit Adams-Bashforth methods to (3), we get

$$f_{i,j}^{n+1} = \tilde{f}_{i,j}^{n} + \Delta t \sum_{k=1}^{s} a_k K_{i,j}^{n,k},$$

where $\tilde{f}_{i,j}^{n} \approx f\left(x_i - v_j\Delta t, v_j, t_n\right)$. In each time level t_{n+1-k}, we are given $K_{i,j}^{n+1-k}$, therefore, we can interpolate $K_{i,j}^{n,k}$ from $K_{\mathcal{I},j}^{n+1-k}$. For the next step, we compute the collision operator $Q_{i,\mathcal{J}}^{n+1}$ from $f_{i,\mathcal{J}}^{n+1}$ using the fast spectral method and modify it by imposing

$$(\Delta v)^{d_v} \sum_j Q_{i,j}^{n+1}\phi_j = 0.$$

Then, save $K_{i,j}^{n+1} = \frac{1}{Kn} Q_{i,j}^{n+1}$.

The general description of semi-Lagrangian based AB methods can be summarized as follows:

Algorithm for AB based SL schemes

1. Using a conservative reconstruction [6], interpolate $\tilde{f}_{i,j}^{n}$ on $x_{i,j}^{n,1}$ from $f_{\mathcal{I},j}^{n}$.
2. For $k = 1, ..., s$, using a conservative reconstruction [6], interpolate $K_{i,j}^{n,k}$ on $x_{i,j}^{n,k}$ from $K_{\mathcal{I},j}^{n}$.
3. Compute numerical solution:

$$f_{i,j}^{n+1} = \tilde{f}_{i,j}^{n} + \Delta t \sum_{k=1}^{s} a_k K_{i,j}^{n,k}.$$

4. Construct $Q_{i,\mathcal{J}}^{n+1}$ using $f_{i,\mathcal{J}}^{n+1}$ using the fast spectral method described in Section 3.
5. For each i, using the weighted L^2-minimization procedure described in Section 5, we modify $Q_{i,\mathcal{J}}^{n+1}$ so that it satisfies

$$(\Delta v)^{d_v} \sum_j Q_{i,j}^{n+1}\phi_j = 0, \quad \phi_j = 1, v_j, |v_j|^2/2$$

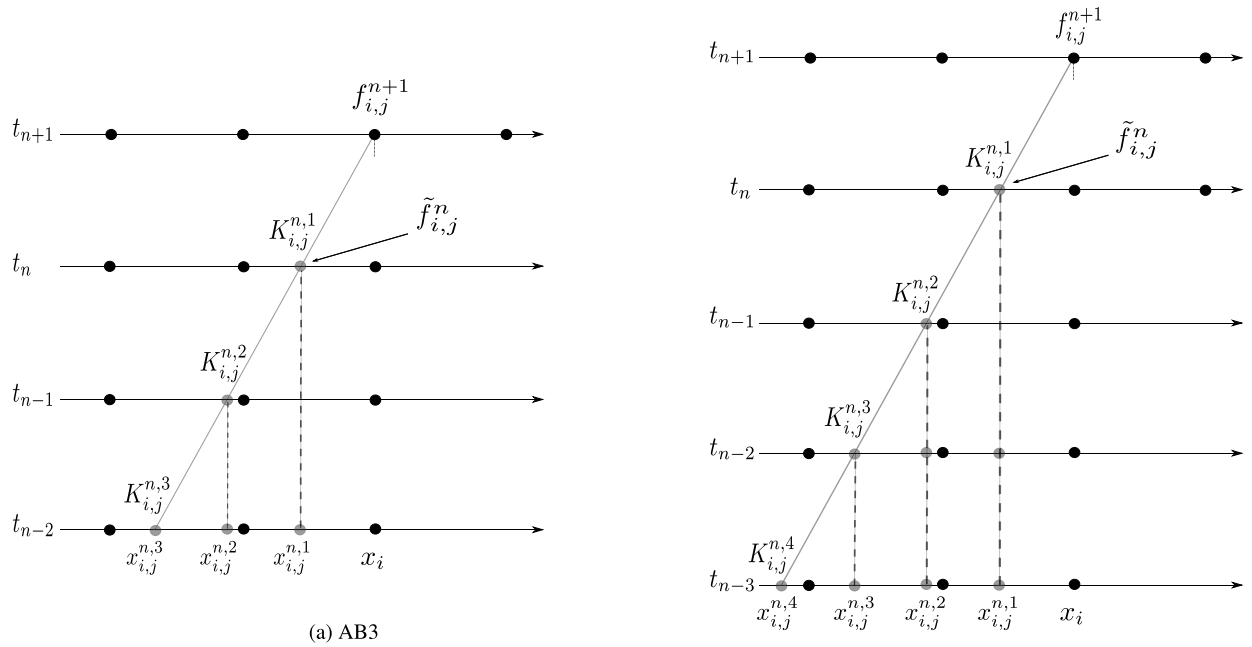
   up to a tolerance (tol).
6. Compute $K_{i,j}^{n+1} = \dfrac{Q_{i,j}^{n+1}}{Kn}$ for each i, j for the next step.

For the iteration, we only need the values of $K_{i,j}^{n+1-k}$, $k = 1, ..., s$ and $f_{i,j}^{n}$, and hence four interpolation procedures and one computation for the collision operator is necessary for each step. In view of this, multi-step methods are more efficient compared to the one-step method. However, we note that these multi-step based SL methods have narrow stability regions compared to RK methods.

AB3: $$f_{i,j}^{n+1} = \tilde{f}_{i,j}^{n} + \Delta t\left(\frac{23}{12}K_{i,j}^{n,1} - \frac{16}{12}K_{i,j}^{n,2} + \frac{5}{12}K_{i,j}^{n,3}\right)$$

AB4: $$f_{i,j}^{n+1} = \tilde{f}_{i,j}^{n} + \Delta t\left(\frac{55}{24}K_{i,j}^{n,1} - \frac{59}{24}K_{i,j}^{n,2} + \frac{37}{24}K_{i,j}^{n,3} + \frac{9}{24}K_{i,j}^{n,4}\right).$$

Remark 2.2. AB based methods require the computation of the collision operator only one time for each time step. Regarding the number of interpolation procedure, the AB3 based SL method involves four interpolations (one for distribution function and three for collision operator), while the AB4 based SL method involves five interpolations (one for distribution function and four for collision operator.)



(a) AB3

(b) AB4

Fig. 4. Illustration of AB3 and AB4 method. The quantities on grey points need interpolation.

3. Spatial reconstruction

In order to implement semi-Lagrangian methods, one has to approximate the distribution function on off-grid points. To this aim, we adopt a high order conservative reconstruction technique [6], which allows us to approximate point-wise values of distribution functions on characteristic feet. The main idea of the method is precomputing the so-called basic reconstruction for each spatial cell, and integrating them to obtain approximation. In [6,7], we choose CWENO polynomials as basic reconstruction. As a consequence, we obtain a high order accurate reconstruction guaranteeing the global conservation of the numerical solutions at the discrete level and preventing spurious oscillations. Below, we briefly describe the conservative reconstruction technique.

Let us consider uniform mesh $x_i = x_L + i\Delta x, i \in \mathcal{I}$ where x_L is the leftmost point of the computational domain, Δx is the size of mesh, $\mathcal{I}$ is the set of indices i. We will denote by I_i the interval that corresponds to the index i: $I_i = [x_i - \Delta x/2, x_i + \Delta x/2]$. Assume that we are given a polynomial $\{R_i\}_{i\in\mathcal{I}}$ defined on each cell I_i. From these, we can construct piecewise polynomials:

$$R(x) = \sum_i R_i(x)\chi_i(x),$$

where $\chi_i(x)$ is the indicator function such that $\chi_i(x) = 1$ for $x \in I_i$ and $\chi_i(x) = 0$ on $x \in I_i^c$. If R_i is a polynomial of degree k, we may have

$$R_i(x) = \sum_{\ell=0}^{k} \frac{R_i^{(\ell)}}{\ell!}(x - x_i)^\ell,$$

where $R_i^{(\ell)}$ is coefficient. Then, for any $\theta \in [0, 1)$, the sliding average of $R(x)$ on the interval $[y_{i-\frac{1}{2}}, y_{i+\frac{1}{2}}] = [x_{i-\frac{1}{2}+\theta}, x_{i+\frac{1}{2}+\theta}]$ (see Fig. 5) gives an approximation of u at $x_{i+\theta} := x_i + \theta\Delta x$ as

$$u(x_{i+\theta}) \approx Q_{i+\theta} = \frac{1}{h}\int_{x_{i-\frac{1}{2}+\theta}}^{x_{i+\frac{1}{2}+\theta}} R(x)dx.$$

Note that $Q_{i+\theta}$ is explicitly computed as follows:

$$Q(x_i + \theta\Delta x) = \sum_{\ell=0}^{k} (\Delta x)^\ell \left(\alpha_\ell(\theta)R_i^{(\ell)} + \beta_\ell(\theta)R_{i+1}^{(\ell)}\right),$$

where

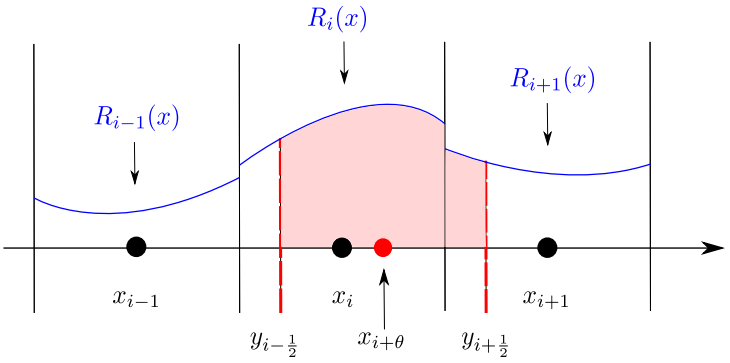


Fig. 5. Illustration of conservative reconstruction taken from Fig. 1 in [6].

$$\alpha_\ell(\theta) = \frac{1-(2\theta-1)^{\ell+1}}{2^{\ell+1}(\ell+1)!}, \quad \beta_\ell(\theta) = \frac{(2\theta-1)^{\ell+1}-(-1)^{\ell+1}}{2^{\ell+1}(\ell+1)!}.$$

For this reason, the most important step of the approach is the decision of the basic reconstruction. In [6], we confirm that CWENO23 [20] and CWENO35 [8] reconstructions could be one good candidate for the choice. Hereafter, we denote the conservative reconstruction based on CWENO polynomials QCWENO. For reader's convenience, in Appendix A we present the explicit form of $R_i(x)$ of QCWENO35 that corresponds to CWENO35 [6]. In the numerical tests, we adopt QCWENO23 and QCWENO35 reconstructions. For a two dimensional test, we use the same two dimensional QCWENO23 reconstruction adopted in [6].

4. Fast spectral method for Boltzmann collision operator

Computing numerical solutions to the Boltzmann equation has been a challenging problem due to the high dimension of its collision integral. For treating this problem, a class of numerical methods based on Fourier spectral approximation has been proposed and studied in numerous paper [17,24,22,11,15,12,19,3,21]. In this section, we review the fast spectral method [22], which is employed in this work.

The spectral method (also called Fourier-Galerkin method) is based on the Fourier series expansion of the velocity distribution function and the Boltzmann collision. Let us consider uniformly distributed frequencies indexed by k, where $k = (k_1, ..., k_{d_v})$ is the multi-index such that $-\frac{N}{2} \le k_1, ..., k_{d_v} \le \frac{N}{2} - 1$. The number N will set to be N_v in our simulation. We assume that the distribution function is well approximated by a function with the support contained in $D_T := [-T, T]^{d_v}$. We then approximate the solution by the following truncated Fourier series [24]:

$$f_N(v) = \sum_{k=-\frac{N}{2}}^{\frac{N}{2}-1} \hat{f}_k e^{i\frac{\pi}{T}k\cdot v}, \quad i = \sqrt{-1}$$

where the summation with respect to k means

$$\sum_{k=-\frac{N}{2}}^{\frac{N}{2}-1} = \sum_{k_1,\dots,k_{d_v}=-\frac{N}{2}}^{\frac{N}{2}-1}$$

and the Fourier mode $\hat{f}_k$ that corresponds to the frequency index k is given by

$$\hat{f}_k = \frac{1}{(2T)^{d_v}} \int_{D_T} f(v) e^{-i\frac{\pi}{T}k\cdot v} dv.$$

More precisely, if a distribution function f is compactly supported on a ball centered at the origin with radius S and the period satisfies the $T \ge \frac{1+3\sqrt{2}}{2} S$, then there is no aliasing error, i.e., the evaluation of the collision operator obtained by the periodized function and the original compactly supported function are the same. For details, we refer to the paper [24,22]. In [24], the explicit form of Fourier mode of collision operator is provided as follows:

$$\hat{Q}_k = \sum_{\ell+m=k,\, k=-\frac{N}{2}}^{\frac{N}{2}-1} \hat{\beta}(\ell, m) \hat{f}_\ell \hat{f}_m, \quad -\frac{N}{2} \le k \le \frac{N}{2} - 1 \tag{5}$$

where $\ell = (\ell_1, ..., \ell_{d_v})$ and $m = (m_1, ..., m_{d_v})$. Here $\hat{\beta}(\ell, m) = \beta(\ell, m) - \beta(m, m)$ is the so-called *kernel mode*. The explicit form of $\beta(\ell, m)$ for the Maxwellian and hard sphere molecules can be found in [24]. If one performs the summation (5) naively, the corresponding

computational cost is $O(N^{2d_v})$. As N and d_v become larger, the resulting spectral method becomes expensive. To overcome this, in [22] authors consider the so-called Carleman representation and approximate the coefficients $\beta(\ell, m)$ with $\beta^f(\ell, m)$:

$$\beta(\ell, m) \approx \beta^f(\ell, m) = \frac{\pi}{M} \sum_{p=0}^{M-1} \alpha_p(\ell)\alpha'_p(m),$$

where

$$\alpha_p(\ell) := \phi_R^2(\ell \cdot e_{\theta_p}), \quad \alpha'_p(m) = \phi_R^2(m \cdot e_{\theta_p + \frac{\pi}{2}})$$

This finally gives us a convolution structure:

$$\hat{Q}_k^f = \frac{\pi}{M} \sum_{p=0}^{M-1} \sum_{m=-\frac{N}{2}}^{\frac{N}{2}-1} \left(\alpha_p(k-m)\hat{f}_{k-m}\alpha'_p(m)\hat{f}_m - \hat{f}_{k-m}\alpha_p(m)\alpha'_p(m)\hat{f}_m \right), \quad -\frac{N}{2} \le k \le \frac{N}{2} - 1$$

Thus, the Fast Fourier transform can be implemented for the computation of $\hat{Q}_k^f$ with reduced computational cost $O(M N^{d_v} \log(N^{d_v}))$. Consequently, we can compute the collision operator in velocity space as follows:

$$Q(v) = \sum_{k=-\frac{N}{2}}^{\frac{N}{2}-1} \hat{Q}_k e^{i\frac{\pi}{T}k \cdot v}.$$

For our simulation, we use MATLAB to perform the Fast Fourier transform and inverse Fourier transform.

5. Weighted L^2-minimization for moment correction

In this section, we recall the weighted L^2-minimization technique [1] which are used for preventing loss of conservation that arises in the computation of the collision operator. This method is more robust than the technique without weight [16] in that it takes into account a weight function which can be helpful to prevent negative values of the distribution function.

To describe the detail, let us consider reference mass, momentum and energy $\mathcal{U} := (\rho, \rho U, E)^\top \in \mathbb{R}^{d_v+2}$, which are physical quantities that we want to preserve throughout numerical simulation. Considering these constraints, we solve a constrained L^2-minimization problem. That is, given an initial guess $Q \in \mathbb{R}^{N_v{}^{d_v}}$ we consider a weight function $w \in \mathbb{R}^{N_v{}^{d_v}}$ and look for a solution $g \in \mathbb{R}^{N_v{}^{d_v}}$ of the following weighted L^2-minimization problem:

$$\min_g \left\| Q \circ w - g \right\|_2^2 \quad \text{s.t} \quad Cg = \mathcal{U} \tag{6}$$

where $\circ$ denotes the componentwise multiplication, i.e.

$$Q \circ w \equiv \left(Q_1 w_1, Q_2 w_2, \cdots, Q_{N_v{}^{d_v}} w_{N_v{}^{d_v}} \right)^\top \in \mathbb{R}^{N_v{}^{d_v}}$$

and

$$Q \equiv (Q_1, Q_2, \cdots, Q_{N_v{}^{d_v}})^\top \in \mathbb{R}^{N_v{}^{d_v}}, \quad g = (g_1, g_2, \cdots, g_{N_v{}^{d_v}})^\top \in \mathbb{R}^{N_v{}^{d_v}}.$$
$$w \equiv (w_1, w_2, \cdots, w_{N_v{}^{d_v}})^\top \in \mathbb{R}^{N_v{}^{d_v}}.$$
$$C := \begin{pmatrix} \frac{1}{w_j}(\Delta v)^{d_v} \\ \frac{1}{w_j} v_j (\Delta v)^{d_v} \\ \frac{1}{w_j} \frac{|v_j|^2}{2} (\Delta v)^{d_v} \end{pmatrix} \in \mathbb{R}^{(d_v+2)\times N_v{}^{d_v}}, \quad \mathcal{U} = (\rho, \rho U, E)^\top \in \mathbb{R}^{(d_v+2)\times 1}.$$

The solution vector g is constructed to be close to $Q \circ w$. Thus, $g \circ \frac{1}{w}$ gives the approximation of Q that reproduces exactly the reference moments $\mathcal{U}$. Here the use of weight $w_j \equiv 1$ for all j leads to the classical L^2-minimization.

As in [1], the solution to the weighted L^2-minimization problem (6) can be obtained by the method of Lagrange multiplier. Denote by $\mathcal{L}(g, \lambda)$ the Lagrangian:

$$\mathcal{L}(g, \lambda) = \|Q \circ w - g\|_2^2 + \lambda^\top (Cg - \mathcal{U}),$$

where λ are the corresponding Lagrangian variables. Finding the gradients of $\mathcal{L}$ with respect to g and λ to be zero at the same time:

$$\begin{array}{lll} \nabla_g \mathcal{L} = 0 & \Leftrightarrow & g = Q \circ w + \frac{1}{2} C^\top \lambda \\ \nabla_\lambda \mathcal{L} = 0 & \Leftrightarrow & Cg = \mathcal{U} \end{array},$$

we can find the stationary points λ:

Table 1
Numerical methods.

	Time discretization	Interpolation	type
RK3+QCWENO23+Euler	RK3	QCWENO23	Eulerian
RK3+QCWENO23+SL			Semi-Lagrangian
RK4+QCWENO23+Euler	RK4	QCWENO35	Eulerian
RK4+QCWENO35+SL			Semi-Lagrangian
AB3+QCWENO23+SL	AB3	QCWENO23	Semi-Lagrangian
AB4+QCWENO35+SL	AB4	QCWENO35	Semi-Lagrangian
RK3+2DQCWENO23+SL	RK3	2DQCWENO23	Semi-Lagrangian

$$\lambda = 2(CC^\top)^{-1}\left(\mathcal{U} - C\left(Q \circ w\right)\right).$$

Since the matrix $CC^\top$ is symmetric and positive definite, it is invertible. Consequently,

$$g \circ \frac{1}{w} = Q + C^\top (CC^\top)^{-1}\left(\mathcal{U} - C\left(Q \circ w\right)\right) \circ \frac{1}{w}.$$

We remark that only a few component-wise division and multiplication of vectors of size ${N_v}^{d_v}$ is required for the weigthed approach with respect to the classical approach [16].

6. Numerical test

Throughout this section, we consider three types of numerical tests: (1) one-dimension in space and two-dimension in velocity with Maxwellian molecules, (2) one-dimension in space and three-dimension in velocity with hard sphere molecules, (3) two-dimension in space and two-dimension in velocity with Maxwellian molecules. In each case, velocity domain is truncated by $[-v_{\max}, v_{\max}]^{d_v}$. When determining the time step, we use the CFL number defined by

$$\text{(1D in space)} \quad \text{CFL} = v_{\max}\frac{\Delta t}{\Delta x},$$

$$\text{(2D in space)} \quad \text{CFL} = v_{\max}\left(\frac{\Delta t}{\Delta x_1} + \frac{\Delta t}{\Delta x_2}\right),$$

where Δx_1 and Δx_2 denote the size of mesh in each space dimension.

The abbreviations of schemes used in numerical tests are reported in Table 1.

6.1. Test 1. Accuracy test and optimal CFL number

In this test, we consider smooth initial data to show that the proposed SL scheme gives the expected convergence rate. Then, we look for the optimal CFL numbers for the method.

6.1.1. Convergence rate

Here we check the accuracy of the proposed SL scheme. To this aim, we take two different initial data as in [12,21]. One is the non-equilibrium:

$$f_0^{neq}(x,v) = \frac{1}{2}\frac{\rho_0(x)}{2\pi T_0(x)}\left(\exp\left(-\frac{(v-U_0(x))^2}{2T_0(x)}\right) + \exp\left(-\frac{(v+U_0(x))^2}{2T_0(x)}\right)\right) \tag{7}$$

which is the sum of two Maxwellians centered on U_0 and $-U_0$, respectively. The other one is the equilibrium initial data:

$$f_0^{eq}(x,v) = \frac{\rho_0(x)}{2\pi T_0(x)}\exp\left(-\frac{|v-U_0|^2}{2T_0(x)}\right). \tag{8}$$

In each case, initial density ρ_0, velocity u_0 and temperature T_0 are given by

$$\rho_0(x) = \frac{2+\sin 2\pi x}{2}, \quad U_0(x) = (0.75, -0.75), \quad T_0(x) = \frac{5+2\cos 2\pi x}{20}.$$

We assume periodic boundary condition in space $x \in [-0.5, 0.5]$ and consider the velocity domain $v \in [-8, 8]$ with $N_v = 32$. The final time is taken by $t = 0.75$ at which solutions remain smooth. We implemented all methods described in Table 1 with the weighted L^2-minimization technique using $N_x = 40, 80, 160, 320$ and the fixed time step Δt that correspond to CFL = 4 or 10. For the comparison with finite difference Eulerian based methods (RK3+WENO23+Euler and RK4+WENO35+Euler), we use the time step determined by CFL = 0.5.

In Tables 2–7, we report the relative L^1-errors and the corresponding convergence rates for RK based methods. It appears that spatial errors are dominant for the Knudsen number $\varepsilon = 0.1$ and 1, which gives order 5, the expected order of convergence of the reconstruction QCWENO35. On the other hand, in Table 6 we can also observe order 4, which is the expected order of convergence for RK4 method. This is because time discretization errors become dominant as we increase the CFL number from 4 to 10. Also, Tables 8–11 show that multi-step based methods give expected convergence rate.

Table 2
Test 1: error and convergence rate for $\varepsilon = 1$. RK3+QCWENO23+SL.

Relative L^1 error and convergence rate of density ρ								
	Non-equilibrium				equilibrium			
	CFL = 4		CFL = 10		CFL = 4		CFL = 10	
$(N_x, 2N_x)$	error	rate	error	rate	error	rate	error	rate
(40, 80)	1.023e-04	2.956	3.905e-05	2.995	1.247e-04	2.892	5.634e-05	2.950
(80, 160)	1.318e-05	2.988	4.900e-06	3.000	1.680e-05	2.978	7.294e-06	2.995
(160, 320)	1.661e-06		6.120e-07		2.132e-06		9.149e-07	

Table 3
Test 1: error and convergence rate for $\varepsilon = 0.1$. RK3+QCWENO23+SL.

Relative L^1 error and convergence rate of density ρ								
	Non-equilibrium				equilibrium			
	CFL = 4		CFL = 10		CFL = 4		CFL = 10	
$(N_x, 2N_x)$	error	rate	error	rate	error	rate	error	rate
(40, 80)	1.575e-05	3.018	5.356e-06	2.796	7.381e-05	2.958	3.562e-05	2.934
(80, 160)	1.945e-06	3.010	7.709e-07	2.919	9.499e-06	2.995	4.660e-06	2.973
(160, 320)	2.415e-07		1.020e-07		1.192e-06		5.936e-07	

Table 4
Test 1: error and convergence rate for CFL = 0.5. RK3+WENO23+Euler.

Relative L^1 error and convergence rate of density ρ								
	Non-equilibrium				equilibrium			
	$\varepsilon = 0.1$		$\varepsilon = 1$		$\varepsilon = 0.1$		$\varepsilon = 1$	
$(N_x, 2N_x)$	error	rate	error	rate	error	rate	error	rate
(40, 80)	2.839e-05	4.664	1.595e-06	4.988	2.217e-05	4.761	9.329e-06	4.914
(80, 160)	1.120e-06	5.144	5.026e-08	5.134	8.176e-07	5.111	3.094e-07	5.069
(160, 320)	3.167e-08		1.431e-09		2.366e-08		9.217e-09	

Table 5
Test 1: error and convergence rate for $\varepsilon = 1$. RK4+QCWENO35+SL.

Relative L^1 error and convergence rate of density ρ								
	Non-equilibrium				equilibrium			
	CFL = 4		CFL = 10		CFL = 4		CFL = 10	
$(N_x, 2N_x)$	error	rate	error	rate	error	rate	error	rate
(40, 80)	3.442e-06	4.876	1.098e-06	4.899	9.372e-06	4.906	4.181e-06	4.916
(80, 160)	1.172e-07	4.955	3.679e-08	4.964	3.127e-07	4.974	1.386e-07	4.971
(160, 320)	3.780e-09		1.179e-09		9.954e-09		4.417e-09	

Table 6
Test 1: error and convergence rate for $\varepsilon = 0.1$. RK4+QCWENO35+SL.

Relative L^1 error and convergence rate of density ρ								
	Non-equilibrium				equilibrium			
	CFL = 4		CFL = 10		CFL = 4		CFL = 10	
$(N_x, 2N_x)$	error	rate	error	rate	error	rate	error	rate
(40, 80)	2.572e-07	4.892	1.684e-07	4.258	3.534e-06	4.955	1.670e-06	4.8122
(80, 160)	8.663e-09	4.958	8.802e-09	4.184	1.139e-07	4.988	5.943e-08	4.867
(160, 320)	2.787e-10		4.844e-10		3.591e-09		2.036e-09	

Table 7
Test 1: error and convergence rate for CFL = 0.5. RK4+WENO35+Euler.

Relative L^1 error and convergence rate of density ρ								
	Non-equilibrium				equilibrium			
	$\varepsilon = 0.1$		$\varepsilon = 1$		$\varepsilon = 0.1$		$\varepsilon = 1$	
$(N_x, 2N_x)$	error	rate	error	rate	error	rate	error	rate
(40, 80)	1.593e-06	4.990	2.834e-05	4.672	9.328e-06	4.915	2.216e-05	4.763
(80, 160)	5.013e-08	5.147	1.112e-06	5.186	3.092e-07	5.071	8.160e-07	5.120
(160, 320)	1.415e-09		3.055e-08		9.196e-09		2.347e-08	

Table 8
Test 1: error and convergence rate for $\varepsilon = 1$. AB3+QCWENO23+SL.

Relative L^1 error and convergence rate of density ρ								
	Non-equilibrium				equilibrium			
	CFL = 4		CFL = 10		CFL = 4		CFL = 10	
$(N_x, 2N_x)$	error	rate	error	rate	error	rate	error	rate
(40, 80)	1.022e-04	2.954	3.184e-05	2.689	1.246e-04	2.892	5.475e-05	2.831
(80, 160)	1.319e-05	2.987	4.937e-06	2.908	1.679e-05	2.978	7.692e-06	2.992
(160, 320)	1.664e-06		6.576e-07		2.130e-06		9.672e-07	

Table 9
Test 1: error and convergence rate for $\varepsilon = 0.1$. AB3+QCWENO23+SL.

Relative L^1 error and convergence rate of density ρ								
	Non-equilibrium				equilibrium			
	CFL = 4		CFL = 10		CFL = 4		CFL = 10	
$(N_x, 2N_x)$	error	rate	error	rate	error	rate	error	rate
(40, 80)	1.256e-05	2.994	6.584e-05	2.832	7.516e-05	2.963	1.465e-04	3.110
(80, 160)	1.577e-06	2.991	9.244e-06	3.107	9.637e-06	2.998	1.697e-05	3.034
(160, 320)	1.983e-07		1.073e-06		1.207e-06		2.073e-06	

Table 10
Test 1: error and convergence rate for $\varepsilon = 1$. AB4+QCWENO35+SL.

Relative L^1 error and convergence rate of density ρ								
	Non-equilibrium				equilibrium			
	CFL = 4		CFL = 10		CFL = 4		CFL = 10	
$(N_x, 2N_x)$	error	rate	error	rate	error	rate	error	rate
(40, 80)	3.424e-06	4.868	8.146e-06	4.033	9.112e-06	4.771	9.647e-06	3.162
(80, 160)	1.173e-07	4.970	4.977e-07	4.068	3.338e-07	4.902	1.078e-06	4.398
(160, 320)	3.741e-09		2.967e-08		1.117e-08		5.116e-08	

Table 11
Test 1: error and convergence rate for $\varepsilon = 0.1$. AB4+QCWENO35+SL.

Relative L^1 error and convergence rate of density ρ								
	Non-equilibrium				equilibrium			
	CFL = 4		CFL = 10		CFL = 4		CFL = 10	
$(N_x, 2N_x)$	error	rate	error	rate	error	rate	error	rate
(40, 80)	1.021e-06	4.188	4.272e-05	4.179	3.739e-06	4.853	5.525e-05	5.001
(80, 160)	5.597e-08	4.137	2.358e-06	4.135	1.294e-07	4.867	1.726e-06	4.294
(160, 320)	3.182e-09		1.342e-07		4.432e-09		8.799e-08	

We observe that SL schemes are more efficient and accurate than the corresponding Eulerian based schemes with the same discretization of parameter Δx, Δv. The improvement of efficiency is due to the possibility of using much larger CFL numbers, therefore reducing the total number of time steps. SL schemes are also more accurate because of the reduced error in the treatment of the transport term, especially for very rarefied flow, where the transport is dominant over collisions.

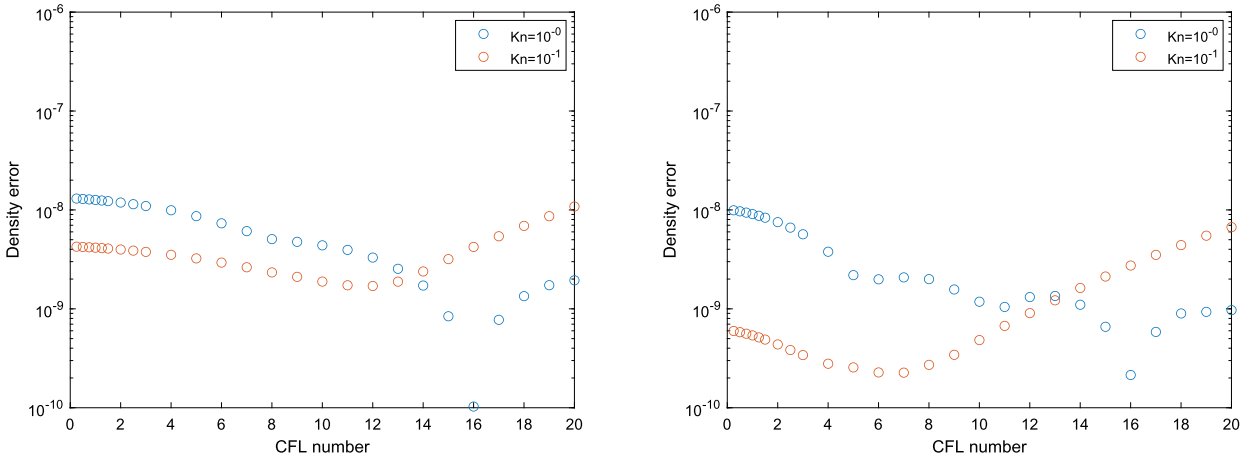
6.1.2. Optimal CFL number

The main advantage of the SL method is the allowance of large time step. Hence, it is important to find the CFL number which gives the smallest errors. Hereafter we call it the optimal CFL number. For this test, we use the same initial data (7) and (8) to compute errors. In Fig. 6, we report the relative L^1-norm of the error of the density obtained by $N_x = 160, 320$. We first look at the case for equilibrium data in Fig. 6a. For a Knudsen number $\varepsilon = 1$, it appears that error becomes very small around CFL$=$ 16. This is because with the choice of $N_v = 32$ and CFL$=$ 16, the feet of the characteristics fall on grid points, so there is no interpolation error, which for other CFL values, is the dominant source of error because for large Knudsen number convection is dominant. For a relatively smaller Knudsen number $\varepsilon = 0.5$, CFL number around 10 gives the smallest error. In Fig. 6b, we can also observe similar tendency with different optimal CFL numbers. We remark that choosing an optimal CFL number depends on the initial data.

6.2. Test 2. A sod problem

In this test, we show that our scheme is able to capture correct behavior of shock profile and it preserves conservative variables. Here we consider initial data as in [21]:

$$(\rho_0, U_{10}, U_{20}, T_0) = \begin{cases} (1, 0, 0, 1), & \text{if} \quad -1 \le x \le 0.5 \\ \left(\frac{1}{8}, 0, 0, \frac{1}{4}\right), & \text{otherwise} \end{cases} . \tag{9}$$



(a) Equilibrium data (b) Non-equilibrium data

Fig. 6. Test 1: CFL number versus error for different Knudsen numbers.

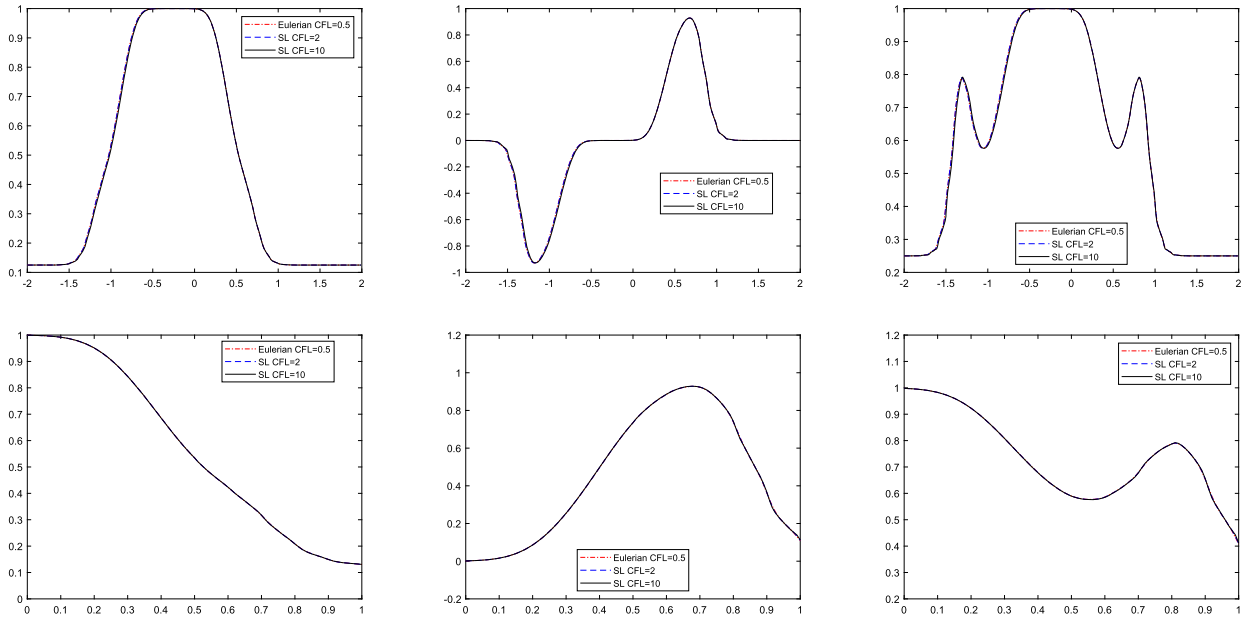


Fig. 7. Test 2: comparison of Eulerian and SL methods for $\epsilon = 0.01$ with L^2-minimization. $N_v = 32$, $v_{\max} = 8$.

In order to check the global conservation, we consider periodic boundary condition on $x \in [-2, 2]$ with $N_x = 400$. To show the effect of L^2-minimization, we perform the test for various discrete velocities with $N_v = 16, 32, 64$ and $v_{\max} = 8$ or 10. We use CFL $= 2$, 10 for the semi-Lagrangian method while we take CFL $= 0.5$ for the Eulerian one. Density, velocity and temperature at time $t = 0.2$ are plotted in Figs. 7-8. Even with the use of larger time steps, the SL based scheme is able to reproduce results which is comparable with those of Eulerian one.

Table 12 shows L^2-minimization can guarantee the conservation of physical quantities up to machine precision even with fewer number velocity grids. However, we should not take too few numbers of velocity grids because it could lead to unstable solutions. Note that even without L^2-minimization, the conservation could be improved if we consider sufficiently large grid numbers in velocity.

6.3. Test 3. Convergence to equilibrium

In [12], the author investigated the behavior of the quantity $\|\rho(t) - \rho_g\|_1$ to estimate the speed of convergence of distribution function to the global equilibrium, and demonstrated that oscillations appear on the difference between global and local equilibria. To reproduce the result, we use the same initial data

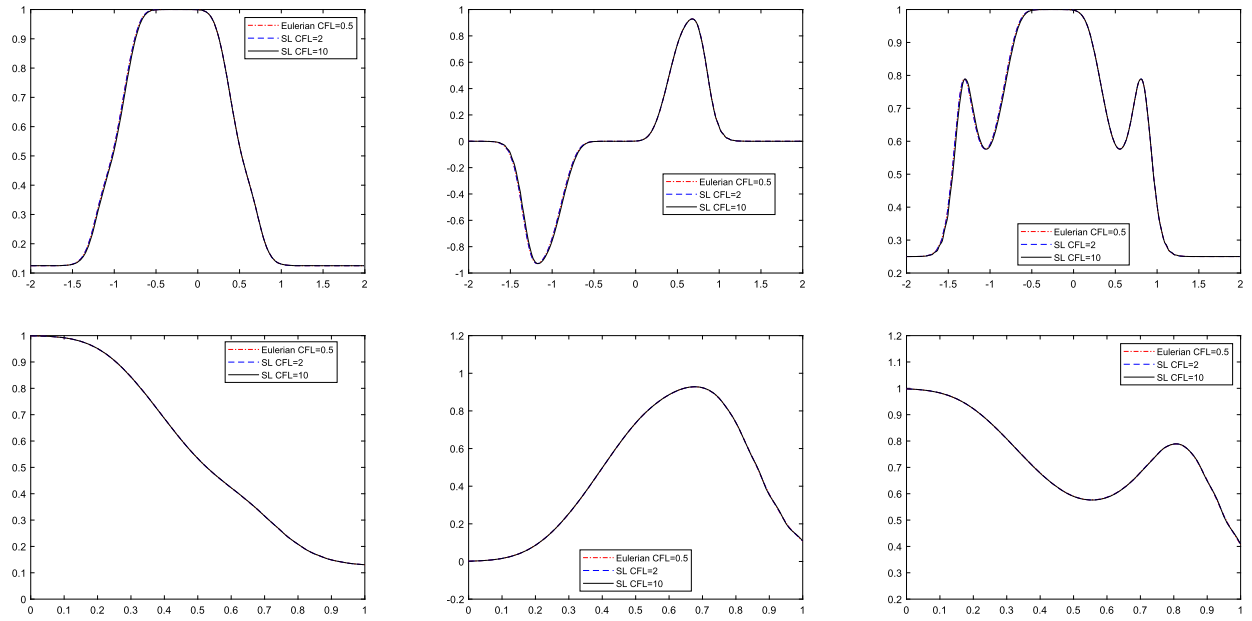


Fig. 8. Test 2: comparison of Eulerian and SL methods for $\varepsilon = 0.01$ with L^2-minimization. $N_v = 64$, $v_{\max} = 10$.

Table 12
Test 2: Conservation error for RK4+QWENO35+SL.

		without L^2-minimization		L^2-minimization	
	N_v	$v_{\max} = 8$	$v_{\max} = 10$	$v_{\max} = 8$	$v_{\max} = 10$
Mass	16	-7.655e-03	-8.710e-02	NaN	NaN
	32	9.493e-08	7.269e-06	-6.096e-15	-9.144e-15
	64	-1.358e-09	-5.628e-09	-7.803e-15	-1.024e-14
Momentum	16	1.152e-03	1.903e-02	NaN	NaN
	32	9.483e-08	2.755e-07	-2.314e-17	6.028e-17
	64	5.533e-08	8.564e-10	2.263e-17	-3.488e-17
Energy	16	-6.587e-02	-1.150e+00	NaN	NaN
	32	-9.281e-06	-1.013e-04	-7.132e-15	-9.090e-15
	64	-5.407e-06	-2.339e-07	-7.132e-15	-1.035e-14

$$f_0(x,v) = \frac{1 + A_0 \sin(\pi x)}{2\pi T_0}\left[\exp\left(-\frac{|v-U_0|^2}{2T_0}\right) + \exp\left(-\frac{|v+U_0|^2}{2T_0}\right)\right]$$
$$A_0 = \frac{1}{2}, \quad T_0 = 0.125, \quad U_0 = \left(\frac{1}{2}, \frac{1}{2}\right) \tag{10}$$

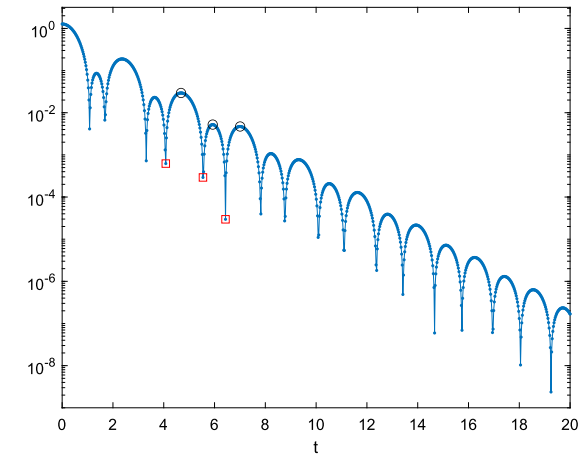
with periodic space domain $x \in [-1,1]$ and velocity domain $v \in [-5,5]$. Here the corresponding global equilibrium is given by

$$\mathcal{M}_g = \frac{\rho_g}{2\pi T_g}\exp\left(-\frac{|v|^2}{2T_g}\right)$$

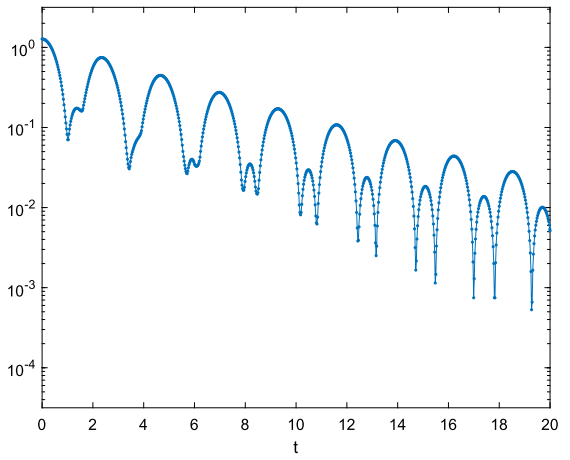
where

$$\rho_g = \int_{-1}^{1}\int_{\mathbb{R}^2} f_0(x,v)\,dv\,dx, \quad T_g = \frac{1}{2\rho_g}\int_{-1}^{1}\int_{\mathbb{R}^2} f_0(x,v)|v|^2\,dv\,dx$$

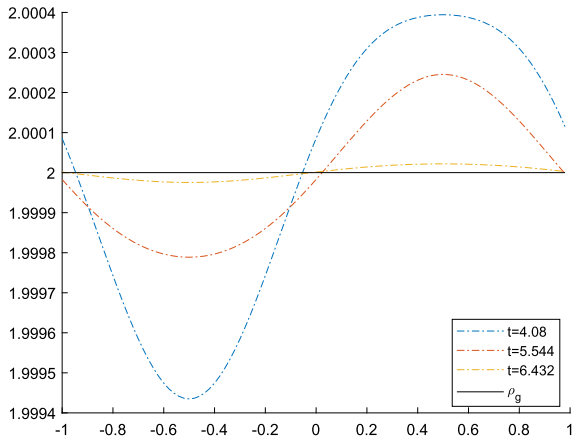
The Knudsen number is taken by $\varepsilon = 0.5, 0.1$ and we set final time by $t = 20$ to observe the long time behavior of numerical solution. Our scheme has advantage in this test because it allows us to use large time step. Here we set Δt that corresponds to CFL $= 6$. In Fig. 9 we show the results, and they are consistent with the literature [13]. Furthermore, in Fig. 10 we can observe the fluctuation of densities around 2 which corresponds to the global equilibrium. For more detailed explanation about the oscillating behavior, we refer to [10].



(a) $\varepsilon = 0.5$



(b) $\varepsilon = 0.1$



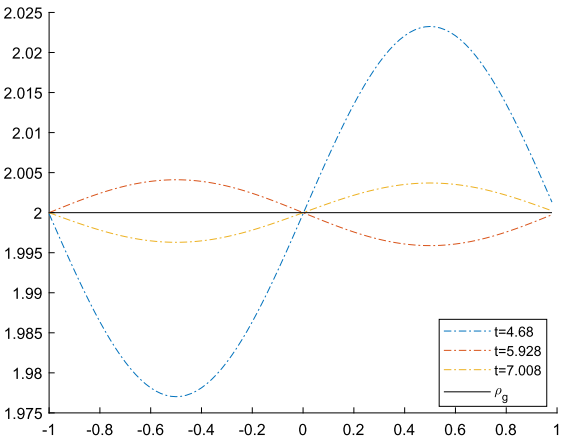


Fig. 9. Test 3: time evolution of $\|\rho - \rho_g\|_1$ associated to initial data (10).

6.4. Test 4: A sod problem with hard sphere molecule in three dimension

Here we solve a Sod problem with three dimensional hard sphere molecules. Initial data is the local Maxwellian with similar macroscopic quantities described in (9), namely,

$$(\rho_0, U_{10}, U_{20}, U_{30}, T_0) = \begin{cases} (1,0,0,0,1), & \text{if} \quad -1 \le x \le 0.5 \\ \left(\frac{1}{8},0,0,0,\frac{1}{4}\right), & \text{otherwise} \end{cases} . \tag{11}$$

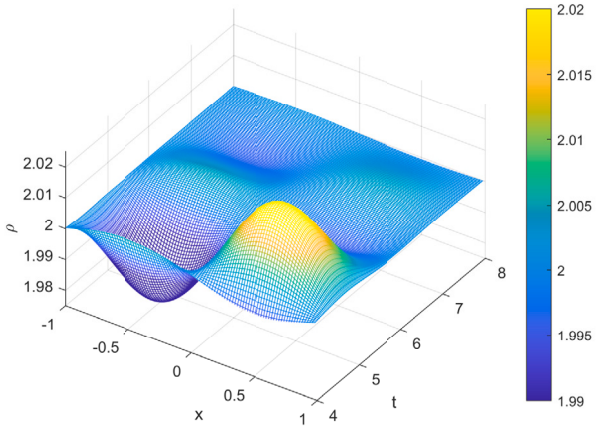
We set periodic boundary condition on the space domain $x \in [-2,2]$ with $N_x = 400$. Here we fix $N_v = 32$ for $v_{\max} = 8$ or $N_v = 64$ for $v_{\max} = 10$. Time step is fixed to satisfy CFL $= 4$. In both cases, we use eight discrete angles in each direction. Density, velocity and temperature at time $t = 0.5$ are plotted in Fig. 11. We see that the two solutions are very similar. Compared to the two-dimensional case in Fig. 8, for $\varepsilon = 0.01$, it appears that in the three dimensional case the width of the shock is narrower than in the two dimensional case.

6.5. Test 5. Two dimensional vortex

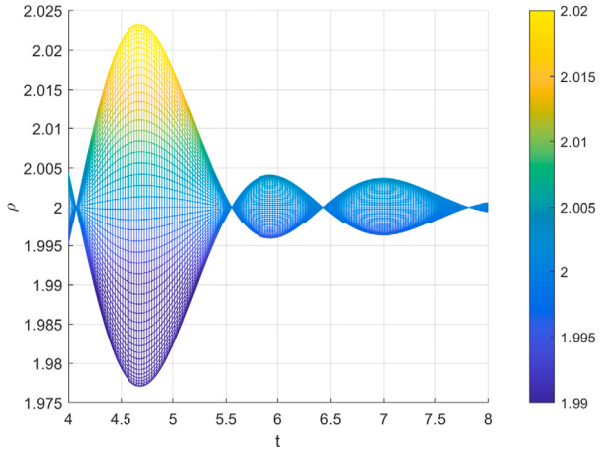
Here we present a numerical result for a two-dimensional problem treated in [15]. In the problem, we observe the behavior of a vortex in the space domain $[0,10]^2$ with periodic boundary conditions. Assuming that the vortex is initially centered at $(5,5)$ with an ambient flow characterized by a gas with density $\rho_\infty = 1$, mean velocity $U_\infty = (1,1)$ and temperature $T_\infty = 1$ respectively, it returns to the center at time $t = 10$. For the initial distribution function, we take local Maxwellian with macroscopic moments:

$$U(x,0) = U_\infty + \delta U, \quad T(x,0) = T_\infty + \delta T$$

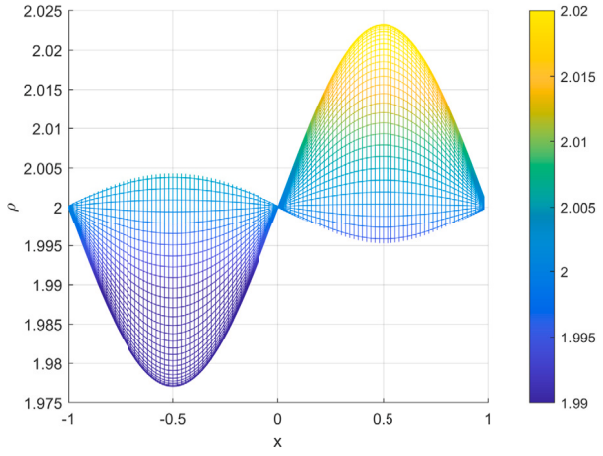
where



(a) $\rho(x,t)$ at $(x,t) \in [-1,1] \times [4,8]$ in Figure 9a



(b) $\rho(x,t)$ at $t \in [4,8]$ in Figure 9a



(c) $\rho(x,t)$ at $x \in [-1,1]$ in Figure 9a

Fig. 10. Test 3: time evolution of $\rho(x,t)$ associated to initial data (10) for $\varepsilon = 0.5$.

Table 13
Test 5: error and convergence rate for $\varepsilon = 0.1$. RK3+2DQCWENO23+SL.

	Relative L^1 error and convergence rate							
	ρ		U_1		U_2		T	
$(N_x, 2N_x)$	error	rate	error	rate	error	rate	error	rate
(25, 50)	1.041e-04	2.996	2.954e-04	2.887	2.975e-04	2.890	8.713e-05	2.946
(50, 100)	1.305e-05	3.009	3.993e-05	2.969	4.014e-05	2.971	1.131e-05	2.992
(100, 200)	1.622e-06		5.099e-06		5.120e-06		1.422e-06	

$$\delta U_1 = -(x_2 - 5)\frac{\beta}{2\pi}\exp\left(\frac{1-r^2}{2}\right), \quad \delta U_2 = (x_1 - 5)\frac{\beta}{2\pi}\exp\left(\frac{1-r^2}{2}\right), \quad \delta T = -\frac{(\gamma-1)\beta}{8\gamma\pi^2}\exp\left(1-r^2\right)$$

with $\gamma = 2$, $\beta = 5$ and $r^2 = (x_1 - 5)^2 + (x_2 - 5)^2$. For density, we use

$$\rho(x,0) = \rho_\infty \left(\frac{T(x,0)}{T_\infty}\right)^{\frac{1}{\gamma-1}}$$

We consider periodic boundary condition in space, and the truncated velocity domain is $[-7.5, 7.5]^2$. Numerical solutions are computed up to the final time $t_f = 10$. In Table 13, we report the accuracy of our method using uniform grids with $N_{x_1} = N_{x_2} = 25, 50, 100, 200$ and $N_v = 32$ for each velocity direction. The time step Δt that correspond to CFL = 1.05 is used. We take eight discrete angles $M = 8$ in all cases. From the table, we confirm the third order accuracy in time and space of the RK3+2DQCWENO23+SL method. In Fig. 12, we describe our numerical solutions for $N_{x_1} = N_{x_2} = 50, 100, 200$. Note that our solution has the cylindrical symmetry and shows good agreement with the numerical results in [15].

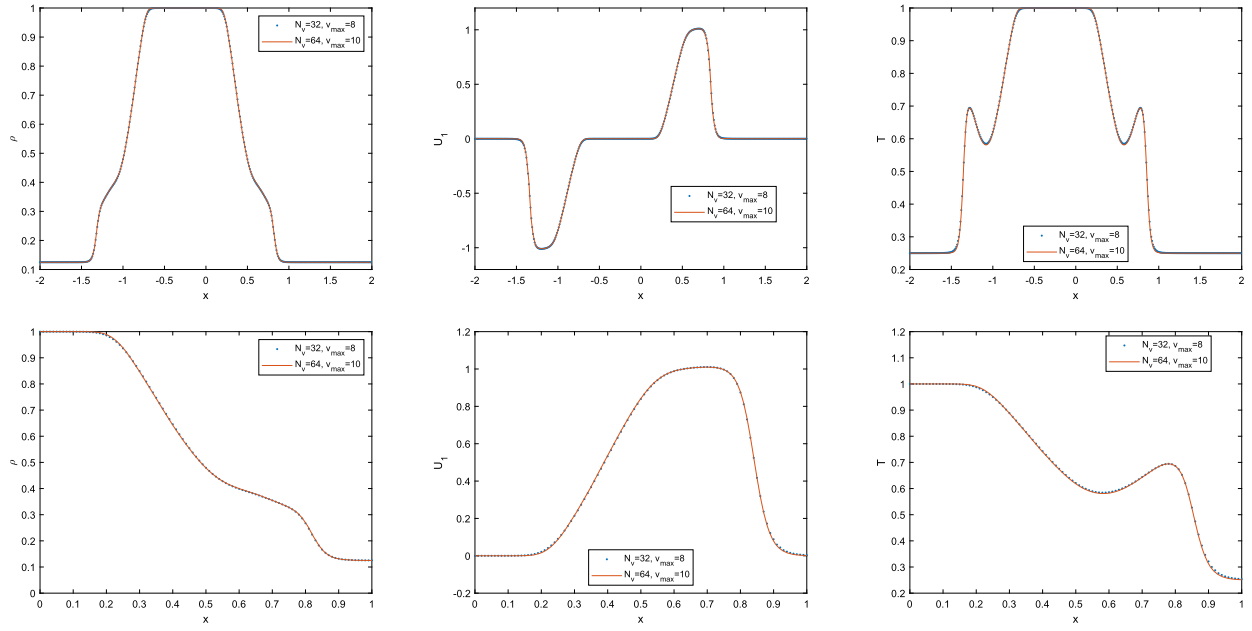


Fig. 11. Test 4: Sod problem with $\epsilon = 0.01$ and CFL$= 4$ for three dimensional hard sphere molecules.

7. Conclusion

In this paper, we propose a SL method that can efficiently solve the Boltzmann equation in the rarefied regime. The main novelty consists in combining the semi-Lagrangian approach for transport with the Fourier-spectral method of Mouhot and Pareschi for the computation of the collision term. This allows the use of larger time step Δt, and provides higher accuracy than the corresponding Eulerian methods. The conservation error associated to the interpolation procedure and caused by the non-linear reconstruction is a major problem associated to semi-Lagrangian method for non-linear problems. It is overcome here by using our conservative reconstruction [6], which avoids such conservation errors. The loss of conservation that comes from the use of the fast spectral method for collision operator is solved using weighted L^2-minimization [1]. High order non-splitting schemes are obtained by adopting Runge-Kutta or Adams-Bashforth methods in time, resulting in overall more efficient methods than the corresponding ones of the same order based on splitting. Finally, although the proposed schemes have shown to be very effective for gas dynamics in rarefied regime (large and moderate Knudsen number), they could be extended to treat the Boltzmann equation with relatively small Knudsen number, by applying a penalization technique similar to the one adopted in [12] in the case of Eulerian methods. This case is also under investigation. It is also possible to extend the conservative SL technique to other kinetic equations, such as, for example, Vlasov-Poisson system (VP) or Landau-Fokker-Plank (LFP) equations. Non-splitting semi-Lagrangian methods for the Vlasov-Poisson system have been studied in [25,29]. However, the extension to these cases is technically complicated for the following reasons: (1) in case of Vlasov equations the trajectories in phase space are not straight lines, therefore the technique we adopt here cannot be used as it is; (2) the LFP operator can be treated with a Fourier spectral method, by adopting the technique proposed in [18], however a different kind of stiffness appears there, i.e. the restriction is of parabolic type in velocity: $\Delta t < \text{CFL}(\Delta v)^2$. That is, one has to take into account the main stability restriction that comes from the operator as well as convection terms. Notwithstanding such difficulties, the use of SL schemes in those more challenging contexts is certainly worth exploring.

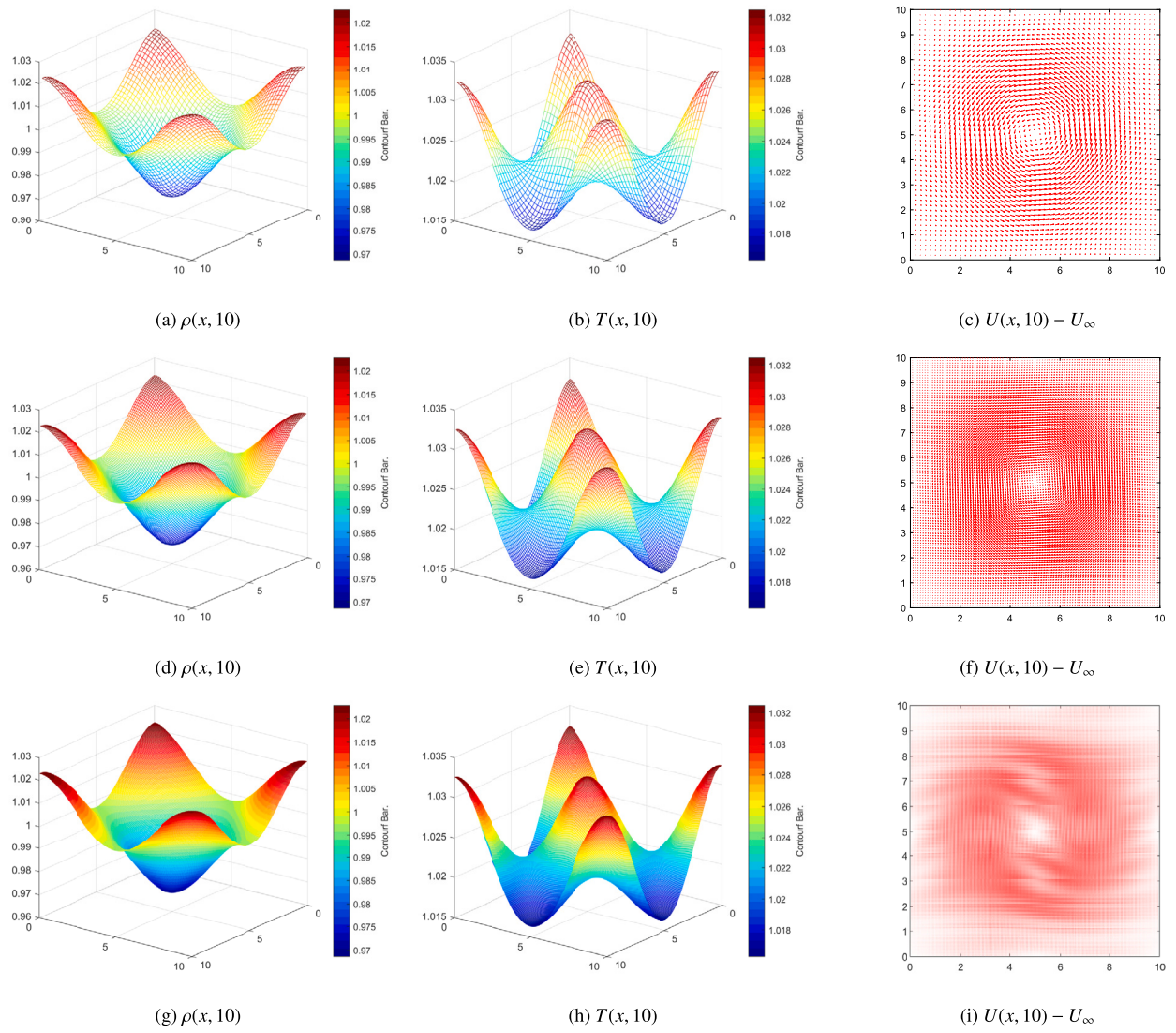
CRediT authorship contribution statement

Sebastiano Boscarino: Methodology, Software. **Seung Yeon Cho:** Methodology, Software. **Giovanni Russo:** Conceptualization, Methodology.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Sebastiano Boscarino reports financial support was provided by the Government of Italy, Ministry of Education, University and Research. Seung Yeon Cho reports financial support was provided by National Research Foundation of Korea. Giovanni Russo reports financial support was provided by the Government of Italy, Ministry of Education, University and Research.



(a) $\rho(x, 10)$ (b) $T(x, 10)$ (c) $U(x, 10) - U_\infty$

(d) $\rho(x, 10)$ (e) $T(x, 10)$ (f) $U(x, 10) - U_\infty$

(g) $\rho(x, 10)$ (h) $T(x, 10)$ (i) $U(x, 10) - U_\infty$

Fig. 12. Test 5: 2D vortex problem. Macroscopic variables at time $t = 10$ for $\epsilon = 0.1$. The first, second and third rows correspond to $N_{x_1} = N_{x_2} = 50, 100, 200$, respectively.

Data availability

No data was used for the research described in the article.

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Appendix A. CWENO35 reconstruction

For the description of CWENO35, the first step is to consider an optimal polynomial $P^i_{OPT}(x)$ that interpolates central stencil $\{\bar{u}_{i-2}, \bar{u}_{i-1}, \bar{u}_i, \bar{u}_{i+1}, \bar{u}_{i+2}\}$ in the sense of cell average:

$$\bar{u}_i = \frac{1}{\Delta x} \int_{x_i-\Delta x/2}^{x_i+\Delta x/2} P^i_{OPT}(x)dx, \quad \forall i \in I$$

This yields a fourth order polynomial:

$$P^i_{OPT}(x) = \tilde{u}_i + \tilde{u}'_i(x-x_i) + \tilde{u}''_i(x-x_i)^2 + \tilde{u}'''_i(x-x_i)^3 + \tilde{u}''''_i(x-x_i)^4$$

where

$$\begin{aligned}
\tilde{u}_i &= \frac{1067}{960}\bar{u}_i - \frac{29}{480}\left(\bar{u}_{i+1}+\bar{u}_{i-1}\right) + \frac{3}{640}\left(\bar{u}_{i+2}+\bar{u}_{i-2}\right)\\
\tilde{u}'_i &= \frac{17}{24}\frac{\bar{u}_{i+1}-\bar{u}_{i-1}}{\Delta x} + \frac{5}{48}\frac{\bar{u}_{i-2}-\bar{u}_{i+2}}{\Delta x}\\
\tilde{u}''_i &= -\frac{1}{16}\frac{\bar{u}_{i-2}+22\bar{u}_i+\bar{u}_{i+2}}{(\Delta x)^2} + \frac{3}{4}\frac{\bar{u}_{i+1}+\bar{u}_{i-1}}{(\Delta x)^2}\\
\tilde{u}'''_i &= -\frac{1}{6}\frac{\bar{u}_{i+1}-\bar{u}_{i-1}}{(\Delta x)^3} - \frac{1}{12}\frac{\bar{u}_{i-2}-\bar{u}_{i+2}}{(\Delta x)^3}\\
\tilde{u}''''_i &= \frac{1}{24}\frac{\bar{u}_{i-2}+6\bar{u}_i+\bar{u}_{i+2}}{(\Delta x)^4} - \frac{1}{6}\frac{\bar{u}_{i+1}+\bar{u}_{i-1}}{(\Delta x)^4}.
\end{aligned}$$

Next, we consider three quadratic polynomials $P_1(x)$, $P_2(x)$, $P_3(x)$ based on stencils $\{\bar{u}_{i-2}, \bar{u}_{i-1}, \bar{u}_i\}$, $\{\bar{u}_{i-1}, \bar{u}_i, \bar{u}_{i+1}\}$, $\{\bar{u}_i, \bar{u}_{i+1}, \bar{u}_{i+2}\}$, respectively:

$$P_\ell(x) = \sum_{k=0}^{2} a_{\ell k}(x-x_i)^k, \quad \ell = 1,2,3.$$

The numbers $a_{\ell k}$, for $1 \le \ell \le 3$ and $0 \le k \le 2$, are given by

$$\begin{aligned}
a_{10} &= \frac{23}{24}\bar{u}_i + \frac{1}{12}\left(\bar{u}_{i-1} - \frac{1}{2}\bar{u}_{i-2}\right), \quad a_{11} = \frac{3\bar{u}_i - 4\bar{u}_{i-1} + \bar{u}_{i-2}}{2\Delta x}\\
a_{12} &= \frac{\bar{u}_i - 2\bar{u}_{i-1} + \bar{u}_{i-2}}{2(\Delta x)^2}, \quad a_{20} = \frac{13}{12}\bar{u}_i - \frac{1}{24}\left(\bar{u}_{i-1}+\bar{u}_{i+1}\right)\\
a_{21} &= \frac{\bar{u}_{i+1}-\bar{u}_{i-1}}{2\Delta x}, \quad a_{22} = \frac{\bar{u}_{i+1} - 2\bar{u}_i + \bar{u}_{i-1}}{2(\Delta x)^2}\\
a_{30} &= \frac{23}{24}\bar{u}_i + \frac{1}{12}\left(\bar{u}_{i+1} - \frac{1}{2}\bar{u}_{i+2}\right), \quad a_{31} = \frac{3\bar{u}_i - 4\bar{u}_{i+1} + \bar{u}_{i+2}}{-2\Delta x}\\
a_{32} &= \frac{\bar{u}_i - 2\bar{u}_{i+1} + \bar{u}_{i+2}}{2(\Delta x)^2}.
\end{aligned}$$

Then, we consider a polynomial $P_C(x)$ which satisfies

$$P^i_C(x) = \frac{1}{C_C}\left(P^i_{OPT}(x) - C_1 P^i_1(x) - C_2 P^i_2(x) - C_3 P^i_3(x)\right), \tag{A.1}$$

where $C_1 = C_3 = \frac{1}{8}$, $C_2 = \frac{1}{4}$ and $C_C = \frac{1}{2}$. Then, we have

$$P_C(x) = \sum_{k=0}^{4} a_{Ck}(x-x_i)^k,$$

where

$$\begin{aligned}
a_{C0} &= \frac{1}{C_C}(\tilde{u}_i - C_1 a_{10} - C_2 a_{20} - C_3 a_{30}), \quad a_{C1} = \frac{1}{C_C}(\tilde{u}'_i - C_1 a_{11} - C_2 a_{21} - C_3 a_{31})\\
a_{C2} &= \frac{1}{C_C}(\tilde{u}''_i - C_1 a_{12} - C_2 a_{22} - C_3 a_3 2), \quad a_{C3} = \frac{1}{C_C}\tilde{u}'''_i, \quad a_{C4} = \frac{1}{C_C}\tilde{u}''''_i
\end{aligned}$$

We compute non-linear weights

$$\omega_k^i = \frac{\alpha_k^i}{\sum_\ell \alpha_\ell^i}, \quad \alpha_k^i = \frac{C_i}{(\epsilon + \beta_k^i)^p}, \quad k, \ell \in \{1, 2, 3, C\} \tag{A.2}$$

with smooth indicators:

$$\beta_k^i = \sum_{l \geq 1} \int_{x_{i-\frac{1}{2}}}^{x_{i+\frac{1}{2}}} \Delta x^{2l-1} (\partial_x^{(l)} P_k^i(x))^2 dx, \quad k \in \{1, 2, 3, C\}.$$

At the end, we obtain the CWENO35 reconstruction:

$$R_i(x) = \sum_k \omega_k^i P_k^i(x), \quad \sum_k \omega_k^i = 1, \quad \omega_k^i \geq 0, \quad k \in \{1, 2, 3, C\}, \tag{A.3}$$

which can be rewritten as

$$R_i(x) = \sum_{\ell=0}^{4} \frac{R_i^{(\ell)}}{\ell!} (x - x_i)^{(\ell)}, \tag{A.4}$$

with

$$\begin{aligned}
R_i^{(0)} &= \omega_C \left(\frac{577}{480}\bar{u}_i - \frac{29}{240}\bar{u}_{i-1} + \frac{19}{960}\bar{u}_{i-2} - \frac{29}{240}\bar{u}_{i+1} + \frac{19}{960}\bar{u}_{i+2} \right) \\
&\quad - \omega_2 \left(\frac{\bar{u}_{i-1} - 26\bar{u}_i + \bar{u}_{i+1}}{24} \right) + \omega_1 \left(\frac{23}{24}\bar{u}_i + \frac{1}{12}\bar{u}_{i-1} - \frac{1}{24}\bar{u}_{i-2} \right) + \omega_3 \left(\frac{23}{24}\bar{u}_i + \frac{1}{12}\bar{u}_{i+1} - \frac{1}{24}\bar{u}_{i+2} \right) \\
R_i^{(1)} &= -\omega_C \frac{8\bar{u}_{i-1} - \bar{u}_{i-2} - 8\bar{u}_{i+1} + \bar{u}_{i+2}}{12\Delta x} + \omega_1 \frac{3\bar{u}_i - 4\bar{u}_{i-1} + \bar{u}_{i-2}}{2\Delta x} - \omega_3 \frac{3\bar{u}_i - 4\bar{u}_{i+1} + \bar{u}_{i+2}}{2\Delta x} - \omega_2 \frac{\bar{u}_{i-1} - \bar{u}_{i+1}}{2\Delta x} \\
R_i^{(2)} &= 2 \left(\omega_1 \frac{\bar{u}_i - 2\bar{u}_{i-1} + \bar{u}_{i-2}}{2(\Delta x)^2} + \omega_2 \frac{\bar{u}_{i-1} - 2\bar{u}_i + \bar{u}_{i+1}}{2(\Delta x)^2} + \omega_3 \frac{\bar{u}_i - 2\bar{u}_{i+1} + \bar{u}_{i+2}}{2(\Delta x)^2} \right) \\
&\quad - 2\omega_C \left(\frac{10\bar{u}_i - 6\bar{u}_{i-1} + \bar{u}_{i-2} - 6\bar{u}_{i+1} + \bar{u}_{i+2}}{4(\Delta x)^2} \right) \\
R_i^{(3)} &= 6\omega_C \left(\frac{\bar{u}_{i-1} - \bar{u}_{i+1}}{3(\Delta x)^3} - \frac{\bar{u}_{i-2} - \bar{u}_{i+2}}{6(\Delta x)^3} \right), \quad R_i^{(4)} = 24\omega_C \left(\frac{\bar{u}_{i-2} + 6\bar{u}_i + \bar{u}_{i+2}}{12(\Delta x)^4} - \frac{\bar{u}_{i-1} + \bar{u}_{i+1}}{3(\Delta x)^4} \right).
\end{aligned} \tag{A.5}$$

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